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Can large language models automate the HAZOP process without human intervention?

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ABSTRACT

This study investigates whether state-of-the-art multimodal large language models (LLMs) can autonomously perform the entire Hazard and Operability Studies (HAZOP) process without human intervention. Four LLMs—GPT4o, GPT4o-mini, LLAMA, and Gemini—were utilized to generate automatic HAZOP worksheets spanning dozens of pages from an identical piping and instrumentation diagram (P&ID) using a standardized prompt. Their outputs were benchmarked against an expert-prepared refence worksheet and evaluated along two key aspects: (1) model performance, measured by similarity and computational cost and (2) HAZOP performance, measured by the validity of generated scenario and the diversity of safeguards. The results showed that all four LLMs achieved high similarity scores to the reference (F1 scores > 86 %). LLAMA was the most cost-efficient ($0.011 per worksheet), while Gemini generated the greatest number of scenarios (34.3 per worksheet) and safeguards (1.79 per deviation). This study presents a structured framework for evaluating LLMs in HAZOP and highlights their potential as assistive tools in the process safety field. However, key limitations were observed. The proportion of semantically valid scenarios remained low (0.19 to 0.37), and safeguards were heavily biased toward procedural measures, indicating limited diversity in risk-mitigation strategies. To enhance the reliability and practicability of LLM-based HAZOP studies, future research should focus on advanced prompt engineering, domain-specific fine-tuning, and improved reasoning capabilities.

1. Introduction

The Hazard and Operability Study (HAZOP) is a widely used qualitative process hazard analysis (PHA) tool. It was first introduced in the late 1960s by Trevor Kletz and Ellis Knowlton at Imperial Chemical Industries in the United Kingdom (Limb, 2009). HAZOP gained global recognition as a systematic and structured methodology for identifying process hazards and operability issues. Its adoption accelerated particularly after the 1974 Flixborough disaster in the UK, which tragically resulted in 28 fatalities and 36 injuries (HSE, 2024; Elhosary and Moselhi, 2024). The publication of the HAZOP guide by the Chemical Industries Association in 1977 further solidified its role as a foundation of process safety management in the chemical process industry (Elhosary and Moselhi 2024).

HAZOP systematically identifies hazards and derives potential adverse events that arise from deviations in intended design (Crawley and Tyler 2015). It is typically conducted in expert-led brainstorming sessions facilitated by a trained leader. The procedure begins by dividing the process system into sections and nodes, where deviations are generated by systematically combining process parameters with guidewords. For each identified deviation, the corresponding causes and consequences are analyzed, and appropriate safeguards are defined to prevent or mitigate the event (Rossing et al., 2010; Othman et al., 2016; Marhavilas et al., 2020). The results are documented in a worksheet, as illustrated in Table 1. A typical worksheet includes the node number, process design intention, process parameters, guidewords, possible causes, consequences, and existing safeguards, and additional suggestions.

Although this methodology is well established, the traditional HAZOP approach is highly resource-intensive, requiring substantial time and domain expertise. In response to these challenges, the 1980s introduced early efforts in computer-aided HAZOP to improve efficiency (Elhosary et al., 2024). However, these systems faced limitations such as inconsistent scenario generation, data incompatibility, and continued reliance on expert judgement, which ultimately restricted their

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Nomenclature

API	Application programming interfaces
BERT	Bidirectional encoder representations from transformers
BiLSTM	Bidirectional long short-term memory
BoW	Bag of words
CSB	U.S. Chemical Safety Board
FRAM	Functional Resonance Analysis Method
GPT	Generative pre-trained transformers
HAZOP	Hazard and operability
HDBSCAN	Hierarchical density-based spatial clustering of applications with noise
HMM	Hidden Markov model
ISD	Inherently safer design
LDA	Latent dirichlet allocation
LLM	Large language model
LSTM	Long short-term memory
ML	Machine learning
MLP	Multi-layer perceptron
NLP	Natural language processing
NRV	Non return valve
P&ID	Piping and instrumentation diagram
PCA	Principal component analysis
PHA	Process hazard analysis
PRV	Pressure relief valve
RNN	Recurrent neural network
RQ	Research question
Seq2Seq	Sequence to sequence
SOP	Standard operating procedures
STPA	System theoretical process analysis
TF-IDF	Term frequency-inverse document frequency
UMAP	Uniform manifold approximation and projection
ViT	Vision transformer
WCSS	Within cluster sum of squares

widespread adoption (Park et al., 2017, Bai et al., 2021).

In the 2000s and 2010s, further advancements emerged through the digitalization of documentation and the development of expert systems and simulation tools (Elhosary et al., 2024). Tools such as DYNHAZ, Petrohazop, and Process Hazard Analysis Suite (PHASUITE) leveraged historical HAZOP data to replicate analyses for similar process and equipment (Zhao et al., 2005, Labovský et al., 2006, Zhao et al., 2009). Simulation-based software like Aspen HYSYS also enabled semi-—automatic generation of incident scenarios and HAZOP reports (Single et al., 2019). These developments reduced manual input and drew greater interest in HAZOP automation. However, despite these innovations, the high cost and required expertise limited their broader implementation (Single et al., 2019, Bai et al., 2021). Although such tools helped bridge the knowledge gap between experts and novice engineers and reduced repetitive worksheet preparation, they ultimately fell short of achieving full automation (de la O Herrera et al., 2018).

In recent years, the emergence of data-driven approaches—particularly those based on Machine Learning (ML) and Natural Language Process (NLP)—has opened new directions for HAZOP automation. These methods extract patterns and insights from large datasets, such as historical HAZOP records. For example, Bai et al., (2021) applied ant colony optimization and a generalized neural network to perform an automated HAZOP study. In this study, the researchers demonstrated the potential for automating HAZOP by reviewing semantic similarity, node semantic distance similarity, and neighbor node similarity. Feng et al., (2021) applied NLP techniques—specifically, Bidirectional Encoder Representations from Transformers (BERT) embedding combined with a Bidirectional Long Short-Term Memory (BiLSTM) network and attention mechanism—to classify scenarios severity. The study provided consistent assessments, particularly beneficial for small- and medium-sized chemical facilities. Similarly, Wang and Gu (2022) used Latent Dirichlet Allocation (LDA) clustering and a Naive Bayes algorithm to extract hazards from over 5,500 HAZOP records. Ekramipooya et al., (2023) extended these methods to classify HAZOP deviations using NLP and ML techniques. Their subsequent work integrated BERT embeddings with dimensionality reduction (i.e., Uniform Manifold Approximation and Projection (UMAP)), clustering (i.e., Hierarchical Density-based Spatial Clustering of Applications with Noise (HDBSCAN), and Multilayer Perceptron (MLP) classifier) to label and predict safeguards based on identified causes and consequences (Ekramipooya et al., 2024).

During these advances, current ML/NLP-based efforts in HAZOP automation remain limited to partial implementation, focusing mainly on deviations and causes (Feng et al., 2021, Ekramipooya et al., 2023, Ekramipooya et al., 2024). These applications have been confined to classification and retrieval from existing databases. A fully automated HAZOP requires comprehensive identification of deviations, causes, consequences, and safeguards. This level of integration has not yet achieved.

Parallel to these ML/NLP advancements, the advent of Large Language Models (LLMs) has significantly transformed natural language understanding and generation in the process safety. LLMs are built upon the transformer architecture—a neural network framework introduced by Vaswani et al., (2017)—that utilizes attention mechanisms to capture long-range dependencies and contextual relationship in text. These models are trained in vast corpora of text and convert language into high-dimensional vector representations by enabling advanced reasoning and contextual understanding. Modern LLMs have evolved far beyond early rule-based and probabilistic models (e.g., n-grams, Hidden Markov Models), as well as earlier deep learning frameworks like Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, which were constrained by sequential processing and

Table 1
Example of a HAZOP Worksheet (Li and Zhao, 2025).

Name: *72 kt/a Sulfur Recovery and Carbon Disulfide Production Project*
Design Intention: *Liquid sulfur from the storage tank is vaporized and mixed with methane-rich natural gas, reaching a feed temperature of approximately 500 °C, and then fed into the carbon disulfide reactor.*

Parameter	Guide Word	Possible Causes	Consequences	Safeguards	Suggestions
Gas feed flow rate of reactor R0301	More	• Upstream feed control valve FV-301 has malfunctioned or is opened excessively • Sudden increase in the flow rate from the main plant	• Reaction rate accelerates, reactor temperature rises excessively, leading to potential explosion • Reaction rate accelerates, reactor temperature rises excessively, leading to potential explosion	• Safety valve SV-301 and rupture disk • High flow alarm FAH-301	• Add a high-flow interlock to shut down feed if flow exceeds safe limits

vanishing gradients (Naseem et al., 2021; Yang et al., 2023). Sequence-to-sequence (Seq2Seq) models using encoder-decoder architectures marked a significant improvement (Sutskever et al., 2014; Bahdanau et al., 2016) but still faced limitations in handling long sequences and computational efficiency. The transformer model overcame these challenges with its multi-head attention mechanism and scalability, becoming the foundation for today's most powerful LLMs.

Recently, there has been growing interest in applying LLMs in various safety system domains. Ni et al. (2024) used LLMs to identify causal factors in chemical incident investigation reports. Similarly, Ray et al. (2023) analyzed aviation safety reports with LLMs to generate incident summaries, identify causal factors, and evaluate responsible parties, thereby suggesting the feasibility of delegating core analytical tasks normally performed by safety analysts. Other studies have improved LLM reliability by integrating domain-specific knowledge graphs. For instance, Chen et al. (2024) combined an aviation incident causation knowledge graph with LLMs, improving both the efficiency and accuracy of incident analysis, thereby presenting a successful example of coupling big data with LLMs. More recently, Zheng et al. (2025) applied a chemical process safety knowledge graph with LLMs to generate comprehensive emergency response plans for chemical incidents.

Furthermore, LLMs have been actively explored in risk assessment methodologies. Clegg et al. (2024) demonstrated that LLMs can automatically perform failure logic analysis for incident scenarios such as gas leaks and tank overfilling and visualize the results as fault trees, thereby highlighting their potential to enhance safety analysis productivity. Collier et al. (2024) applied LLMs to Failure Modes and Effects Analysis (FMEA) for consumer products and compared the outputs with expert evaluations. Their findings suggest that LLMs can assist brainstorming by generating a broad set of failure modes. Similarly, Diemert and Weber (2023) and Charalampidou et al. (2024) utilized LLMs in System-Theoretic Process Analysis (STPA) to identify unsafe control actions and causal scenarios. Kaya et al. (2025) and Hong et al. (2025) extended this application to full STPA automation. Qi et al. (2025) further investigated an interactive approach, enabling experts and LLMs to jointly identify unsafe control actions during STPA. Collectively, these studies confirm that LLMs can efficiently generate numerous hazards and control measures with reduced cost and time. In addition, LLMs has also been evaluated in the Functional Resonance Analysis Method (FRAM). Kaya et al. (2025) and Sujan et al. (2025) reported that LLMs provide a large volume of useful information quickly. Sujan's study notably incorporated prompts that required self-critique from multiple perspectives, leading the LLM to identify viewpoints often overlooked by human analysts.

Across these studies, LLMs were consistently positioned as supportive rather than fully autonomous tools. Researchers highlighted constraints in consistency and reliability, making human oversight indispensable and preventing LLMs from independently conducting comprehensive risk assessments (Diemert and Weber, 2023; Charalampidou et al., 2024; Kaya, et al., 2025; Hong, et al., 2025; Collier et al., 2024; Sujan, et al., 2025). Consequently, the effectiveness of LLM-assisted risk assessments relies on rigorous expert oversight and iterative human–AI interaction to ensure reliable outcomes (Qi, et al., 2025).

To date, no studies have investigated the applicability of LLMs within traditional hazard identification methodologies such as HAZOP. Improving HAZOP performance is not merely a technical challenge. It is critical for effective risk-based decision-making. As Park et al. (2021) noted, many catastrophic chemical incidents investigated by the U.S. Chemical Safety Board (CSB) involved failures in hazard identification or inadequate safeguard implementation. These findings highlight that accuracy and comprehensiveness of HAZOP outcomes are essential not only for compliance but for preventing incidents and fostering inherently safer design. Enhancing the automation and quality of HAZOP, therefore, has direct implications for both HAZOP operational efficiency and safety reduction outcomes.

In response to these challenges, this study investigates whether LLMs can autonomously execute the entire HAZOP procedure without human intervention and whether the resulting outputs can support effective risk-based decision-making. Within this context, the study addresses the following research questions (RQ):

- **RQ1(Feasibility).** Can current LLMs perform the entire HAZOP procedure autonomously?
- **RQ2(Cost Efficiency).** Does LLM-based HAZOP provide advantages in terms of time and cost compared to traditional methods?
- **RQ3(Practical applicability).** Is the quality of HAZOP results generated by LLMs sufficient for practical application?
- **RQ4(Role).** What role should current LLMs play in HAZOP—full automation or supportive tool—and what are the implications for future development?

The remainder of this paper is organized as follows: Section 2 describes the methodology, including the HAZOP workflow, LLM configuration, and evaluation metrics. Section 3 presents the results of LLM-generated HAZOP performances based on the selected metrics. Section 4 addresses the research questions, discusses threats to validity, and highlights broader implications. Finally, Section 5 concludes the paper and outlines directions for future research.

2. Methodology

This study investigates whether LLMs can automatically perform a complete HAZOP analysis. Four leading LLMs released as of December 2024 are evaluated: (1) OpenAI's GPT4o (OpenAI Platform, 2025a), (2) GPT4o-mini (OpenAI Platform, 2025b), (3) Meta's LLAMA (LLAMA, 2025), and (4) Google's Gemini (DeepMind, 2025).

Each model is tasked with performing a complete HAZOP study based on an identical P&ID. The results are benchmarked against a reference HAZOP worksheet created by domain experts. Two evaluation dimensions are used: model performance (measured by similarity and cost) and HAZOP performance (measured by scenario and safeguard quality). All tasks are executed on a high-performance server equipped with an NVIDIA GeForce RTX 3090 GPU, an Intel(R) Xeon(R) Gold 6326 CPU, and 128 GB of RAM.

As shown in Fig. 1, this methodology consists of three primary steps:

Step 1. HAZOP Generation via LLMs—A single P&ID is provided to all LLMs under identical conditions. Each model generates 30 HAZOP worksheets for consistency in comparison.

Step 2. Word Embedding—The generated worksheets, written in natural language, are converted into numerical vectors using NLP embedding techniques to enable quantitative evaluation.

Step 3. Evaluation—A total of 120 HAZOP worksheets (30 per LLM) are compared against the expert reference across two evaluation dimensions, detailed in Section 2.3.

2.1. Step 1. HAZOP generation via LLMs

Traditionally, HAZOP is a structured, expert-driven process that systematically identifies process deviations, their causes, consequences, and safeguards. In this study, the entire HAZOP procedure is automated using LLMs. A prompt containing a P&ID, process description, and HAZOP guidelines (see Appendix A) is provided to each model. As illustrated in Fig. 2, the workflow includes node identification from the P&ID, generation of incident scenarios through possible process deviations, analysis of causes and consequences, and safeguard recommendations. To ensure consistency across models, a standardized and minimalistic prompt is used. This prompt includes only essential input—the P&ID, process description, and basic process data—and is designed to produce HAZOP worksheets in a uniform format. This uniformity enabling fair and meaningful comparisons across different LLMs.

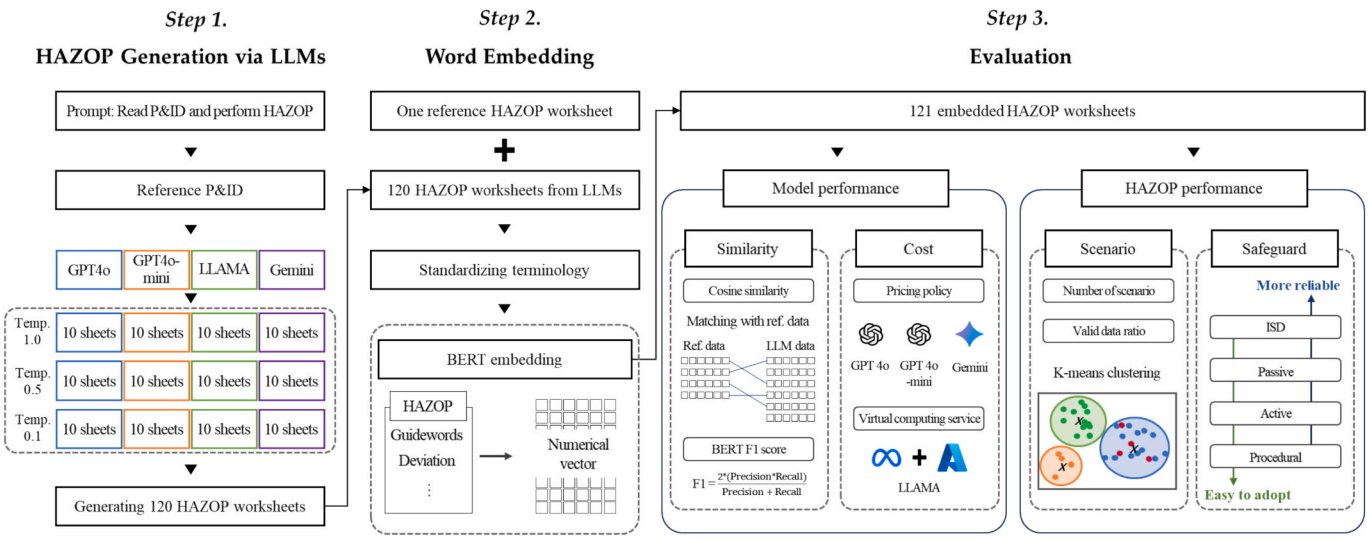


Fig. 1. Overview of Methodology for LLM-based HAZOP Generation and Evaluation.

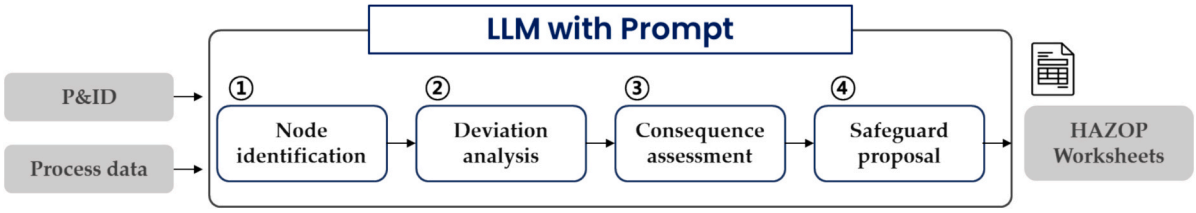


Fig. 2. LLM-based HAZOP workflow for Automated HAZOP Worksheet Generation.

2.1.1. Selected LLM models

This study focuses on four prominent models: GPT4o, GPT4o-mini, LLAMA 3.2, and Gemini 2.0. These models represent the latest advancements in language and vision capabilities. They are tested to investigate the feasibility of fully automating the HAZOP process.

2.1.2. Reference HAZOP worksheet

To evaluate HAZOP performance, a reference P&ID and its corresponding expert-generated HAZOP worksheet are used. Both are sourced from "*HAZOP: Guide to Best Practice (Guidelines to Best Practice for the Process and Chemical Industries)*". This reference provides a well-established example of real-world HAZOP execution and serves as a benchmark for comparison (Crawley and Tyler, 2015). The selected P&ID is shown in Fig. 3, and additional process details, including the full reference worksheet, are provided in Appendix A.

2.1.3. LLM configuration

All four LLMs are guided by a standardized prompt to ensure consistency in data generation. The prompt incorporates three reference elements: (1) a process description, (2) operating procedures, and (3) HAZOP guidelines—all extracted from *HAZOP: Guide to Best Practice* (Crawley and Tyler, 2015). Each LLM is instructed to generate HAZOP worksheets in a consistent tabular format with five columns: Guideword, Effect, Causes, Consequence, and Safeguard. This structured format enables systematic comparison across models.

The exact prompt used is as follows:

"Prompt: Read the P&ID image, perform the HAZOP by referring to the process diagrams in A.1 and A.2, and the HAZOP guidelines in A.3. Then prepare a table-style HAZOP Sheet for each node as presented in A.3. The HAZOP sheet should be written in the order of Guideword/Effect/Cause/Consequence/Safeguard."

In addition to the prompt, the critical hyperparameter—temperature—is used to regulate the level of randomness in the LLM outputs. As formalized by Ackley et al (1985), this parameter shapes probability distribution of model output by redistributing probability mass across possible outcomes. A higher temperature flattens the distribution, increasing the likelihood of less probable outcomes and producing more diverse, creative responses. A lower temperature sharpens the distribution, concentrating probability mass on high-likelihood outputs and yielding more deterministic behavior (Peeperkorn et al., 2024).

To assess the impact of output randomness and creativity, each LLM generates HAZOP worksheets 10 times per temperature setting across three temperature values (0.1, 0.5, and 1.0), resulting in a total of 120 worksheets.

2.2. Step 2. Word embedding

The 120 HAZOP worksheets generated in Step 1 follow a consistent tabular structure—Guideword, Deviation, Causes, Consequence, and Safeguard. However, because the worksheets are written in natural language, they cannot be directly analyzed using numerical methods. To enable computational analysis, the texts are first preprocessed and then transformed into numerical vectors through word embedding techniques.

2.2.1. Standardization of terms

Given the diversity and ambiguity in HAZOP documentation, terminology standardization is performed prior to embedding. Synonyms and variant expressions across worksheets are replaced with standardized terms based on the reference worksheet to ensure uniformity and minimize inconsistencies during evaluation. Table 2 lists representative examples of this preprocessing step for this case study.

2.2.2. Embedding with context-aware language models

To enable quantitative evaluation, each HAZOP worksheet is

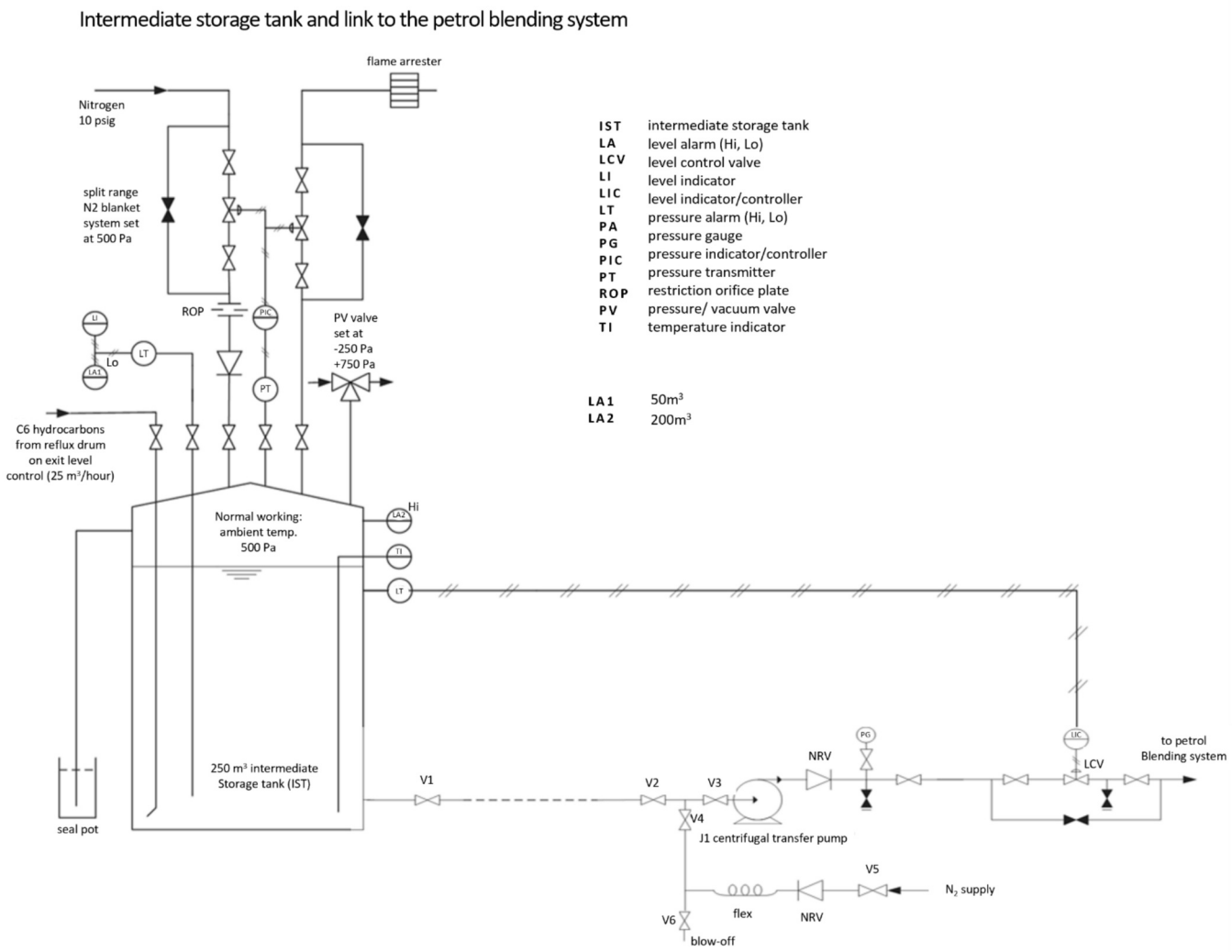


Fig. 3. The Reference P&ID (Crawley and Tyler 2015).

Table 2
Examples of Term Standardization Prior to Embedding.

Before modification	After modification	Before modification	After modification
Nitrogen	N_2	Over-pressurization	Overpressure
Pressure relief valve	PRV	Hydrocarbon	C6
Operator error	Human factor	Human error	Human factor

transformed into a machine-readable format. This is accomplished through word embedding, which converts natural language into numerical vectors while preserving semantic and contextual meaning. A variety of embedding techniques are available, such as Term Frequency-Inverse document frequency (TF-IDF), Bag of Word (BOW), Word2Vec, and BERT.

This study adopted BERT, a transformer-based model recognized for its superior performance in technical and context-rich applications. In this step, each word (or token) in the worksheets is transformed into a numerical vector using BERT embeddings. BERT captures context-sensitive semantics—in other words, it generates different vector representations for the same word depending on surrounding context (Devlin et al., 2019; Zhang et al., 2020). Pre-trained on massive text corpora such as Wikipedia and BooksCorpus, BERT uses a masked language model and bidirectional attention to capture the context in both forward and backward directions (Devlin et al., 2019; Feng et al., 2021). These capabilities make it particularly suitable for technical documents such as HAZOP worksheets, which contain specialized terminology and varied phrasing. Its adaptability and contextual sensitivity have led to widespread adoption in NLP, supported by open-source platforms like Hugging Face (HuggingFace, 2025), which provides pre-trained models, application programming interfaces (API), and tools for custom tools. In this study, the BERT-based embeddings generated in Step 2 form the foundation for Step 3, where worksheet similarity and performance are quantitively evaluated.

2.3. Step 3. Evaluation

The performance of LLM-based HAZOP studies is evaluated using two-main dimensions: (1) model performance and (2) HAZOP performance.

2.3.1. Model performance evaluation

Model performance is assessed in terms of structural similarity to the reference HAZOP worksheets and the cost of generating a single HAZOP worksheet. These two measures provide an overall indication of each LLM's efficiency and cost-effectiveness.

2.3.2. Model performance metric 1: similarity

To evaluate textual similarity between the expert-prepared reference worksheet and each LLM-generated HAZOP worksheet, a three-step process is adopted: (1) embedding (Step 2), (2) row alignment based on contextual similarity, and (3) similarity measurement between aligned content (Fig. 4).

Row alignment is performed to match semantically equivalent rows between the reference and LLM-generated worksheets. This step is essential because LLM-generated worksheets may present the same content in a different row order. Cosine similarity is used to identify the most semantically similar row in the generated worksheet for each row in the reference.

Cosine similarity measures the angle (θ) between two vectors and is calculated as shown in Eqs. (1) and (2) (Lahitani et al., 2016):

$$\text{Cosine similarity} = \frac{1+\cos(\theta)}{2} \tag{1}$$

$$cos(\theta) = \frac{A \bullet B}{\|A\|\|B\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}} \tag{2}$$

here, A and B represent the BERT-embedded vectors of the two compared sentences.

To provide a more fine-grained evaluation, this study employs BERT-based precision, recall, and F1 metrics, which have been widely validated against human references across various domains (Bui et al., 2025; Ray et al., 2023; Zheng et al., 2025). These metrics provide a fine-grained evaluation by comparing individual tokens within the reference and generated worksheets, thereby reflecting nuanced semantic and contextual alignment.

- BERT Recall (R_{BERT}): measures how many of the reference sentence tokens are well-represented in the generated sentence.
- BERT Precision (P_{BERT}): measures how many of the generated tokens are relevant to the reference sentences.
- BERT F1 Score (F_{BERT}): harmonic means of precision and recall, balancing both perspectives (Zhang et al., 2020).

Among these, the BERT F1 Score is adopted as the primary metric because it synthesizes both precision and recall, providing a more robust measure of overall similarity (Zhang et al., 2020).

These values are computed using the following equations:

$$R_{BERT} = \frac{1}{|x|}\sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} x_i^{\top} \hat{x}_j \tag{3}$$

$$P_{BERT} = \frac{1}{|\hat{x}|}\sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} x_i^{\top} \hat{x}_j \tag{4}$$

$$F_{BERT} = \frac{2(P_{BERT} \bullet R_{BERT})}{P_{BERT} + R_{BERT}} \tag{5}$$

here, x represents the set of tokens in the reference sentence, $\hat{x}$ represents the set of tokens in the generated sentence, and $x^T x_i$ is the cosine similarity between individual tokens.

2.3.3. Model performance metric 2: cost

The cost analysis examines the expenses of generating a single HAZOP worksheet, considering input tokens, output tokens, and image processing requirements. This includes API license fees, hardware usage, and other infrastructure expenses.

The pricing structures for the selected LLMs vary with licensing policies, hardware demands, and computational capabilities. For GPT4o and GPT4o-mini (OpenAI), costs are determined by OpenAI's published pricing policy, which employs tiered rates based on token usage. This structure accounts for both prompt and completion tokens, emphasizing the need for optimized token usage to ensure cost efficiency. For LLAMA 3.2 90B (Meta), while the API license is freely available, the model's heavy computational requirements necessitate high-performance hardware. To ensure consistency across the analysis, hardware-related costs are standardized using the pricing models of cloud-based virtual computing services. Specifically, the cost estimation for LLAMA 3.2 is based on Microsoft's Azure AI studio service (Azure Marketplace, 2025), which offers GPU-accelerated instances suitable for processing the model's 90 billion parameters. For Google's Gemini 2.0, the cost assessment is based on Google AI's pricing policy. In contrast, the cost of Google's Gemini 2.0′s is based on Google AI's pricing policy. Its multimodal capabilities, encompassing both text and image processing, introduce additional pricing complexity, as costs depend on the combined computational demands of input and output processing.

2.3.4. HAZOP performance evaluation

HAZOP performance is assessed based on two core objectives: (1) identifying potential incident scenarios and (2) recommending appropriate safeguards.

2.3.4.1. HAZOP performance metric 1: scenario. The first performance metric evaluates the ability of LLMs to generate potential incident scenarios from a given P&ID. This evaluation is grounded in two quantitative indicators: the total number of scenarios generated and the valid scenario ratio. The total number of scenarios represents the volume of HAZOP scenarios generated across 30 repeated runs with a consistent prompt. This indicator captures the model's capacity to identify diverse process deviations and hazards, thereby highlighting its generative comprehensiveness. In contrast, the valid scenario ratio measures the semantic relevance and coherence of generated scenarios by comparing them with expert-generated reference data. A structured three-step

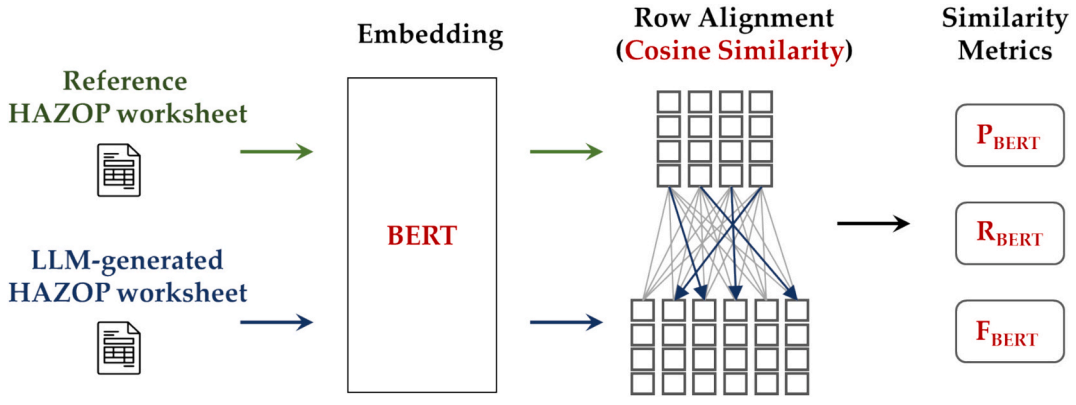


Fig. 4. Workflow of the Similarity Evaluation.

analysis framework is adopted, as illustrated in Fig. 5.

In the first step, the high-dimensional semantic vectors of the BERT-Embedded are reduced using Principal Component Analysis (PCA). PCA projects the data into two principal components, preserving the most significant variance while simplifying the structure for visualization and clustering. This dimensionality reduction facilitates better interpretability without substantial loss of semantic information (Zhang et al, 2023, Ji et al., 2021).

In the second step, K-means clustering is performed on the PCA-reduced embeddings to uncover latent groupings based on semantic similarity. As an unsupervised machine learning algorithm, K-means partitions unlabeled data into K distinct clusters by minimizing intra-cluster variance using Euclidean distance (Wagstaff et al., 2001). The clustering objective is defined by the Within-Cluster Sum of Squares (WCSS), or cost function (J), given by Eq. (6) (Liu et al., 2021):

$$J = \sum_{j=1}^{k} \sum_{i=1}^{n} \left\| m_j - x_i^{(j)} \right\|^2 \tag{6}$$

where, J is the cost function, k is the number of clusters, n is the number of data points in cluster j, m_j is the center of cluster j, and $x_i^{(j)}$ represents the data points of cluster j.

Although increasing k generally lowers WCSS, setting K too high can lead to overfitting and fragmented clusters. To balance cluster granularity and cohesion, the elbow method is used to identify the optimal number of clusters for each model. This method locates the point at which additional clusters yield diminishing returns in reducing WCSS (Bholowalia and Kumar, 2014).

In the third step, the valid scenario ratio is calculated to quantify how many LLM-generated scenarios fall into the same cluster as expert reference. After identifying the cluster containing the majority of expert scenarios, the valid scenario ratio is computed using Eq. (7):

$$\textit{Valid scenario ratio} = \frac{|C_{Reference}|}{\textit{Number of total scenarios}} \tag{7}$$

here, $C_{Reference}$ denotes the number of LLM-generated data points within the reference cluster. This indicator captures the proportion of semantically valid outputs, providing a robust indicator of alignment between model-generated scenarios and expert knowledge.

2.3.4.2. HAZOP performance metric 2: safeguard. Beyond scenarios identification, a fundamental goal of HAZOP is to recommend safeguards to effectively mitigate risks to personnel, equipment, the environment, and surrounding communities (Chastain et al, 2017; Park et al, 2018; Almousa et al, 2025). Accordingly, this study evaluates the quality, diversity, and practical applicability of safeguards proposed by LLM-based HAZOP. Safeguards are generally categorized as follows (CCPS, 1998):

Inherently Safer Design (ISD) / Passive- ISD strategies aim to eliminate or significantly reduce hazards at the source through process design choices. Common ISD principles include minimization, substitution, moderation, and simplification. Passive safeguards, although distinct from ISD, align with its philosophy by mitigating risks through physical design elements that operate without human input or active instrumentation—e.g., rupture disks, blast walls, or double-walled containment.

Active- Active safeguards function by detecting abnormal conditions and initiating automatic responses. These include sensors, alarms, emergency shutdown systems, control valves, and interlocks. Although effective, they depend on instrumentation, calibration, and logic systems, which introduce complexity and maintenance requirements.

Procedural- Procedural safeguards rely on administrative controls, such as standard operating procedures (SOPs), training programs, and communication protocols. Although they are often easier to implement, they are the most vulnerable to human error and provide the least inherent reliability.

Each safeguard plays an important role in risk management. However, as shown in the hierarchy of risk management (Fig. 6), ISD and passive measures provide the highest level of reliability, followed by active and procedural safeguards. Notably, while procedural safeguards are the easiest to implement, they are the least robust, highlighting the inverse relationship between ease of adoption and reliability (Baybutt, 2016).

To evaluate the reliability and practicality of safeguards generated by LLMs, this study classifies each safeguard into its corresponding category and sub-index, as outlined in Fig. 7 and Table 3. This classification enables both qualitative and quantitative evaluation of the safeguard diversity and relevance from a process safety engineering

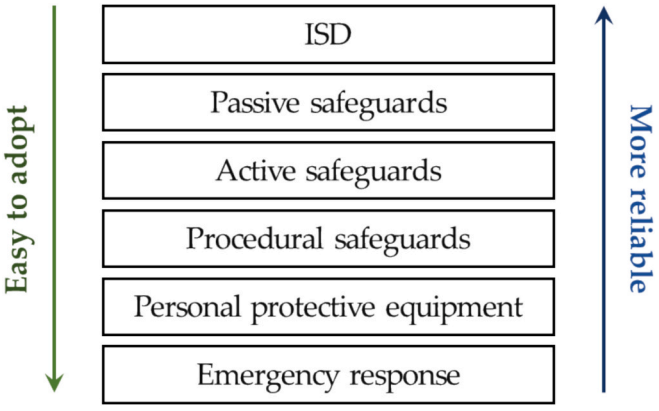


Fig. 6. Hierarchy of Process Risk Management Strategies (Baybutt, 2016).

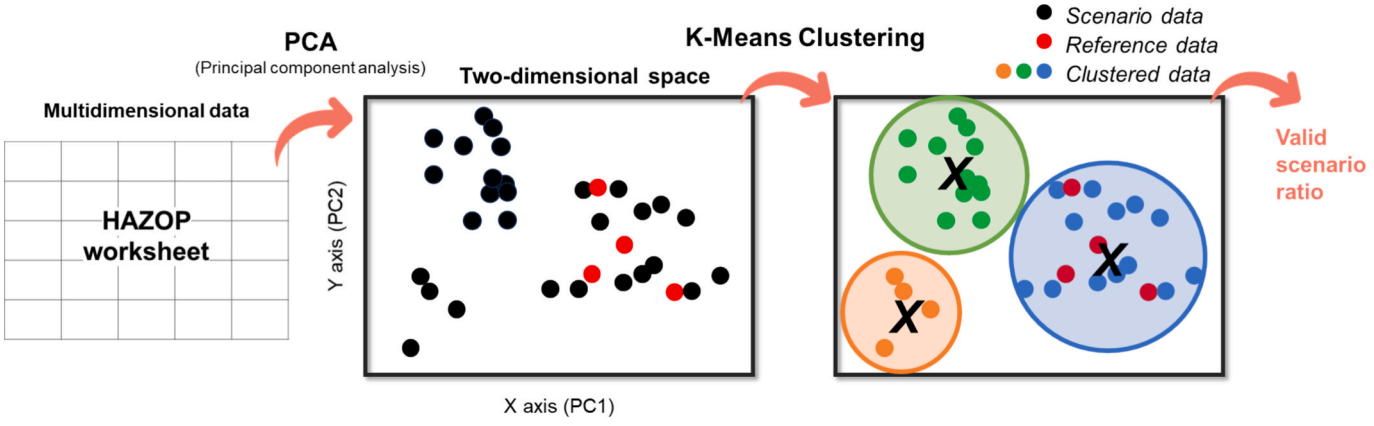


Fig. 5. Three-Step Evaluation of HAZOP Scenario Validity Using PCA, K-means Clustering, and Valid Scenario Ratio Calculation.

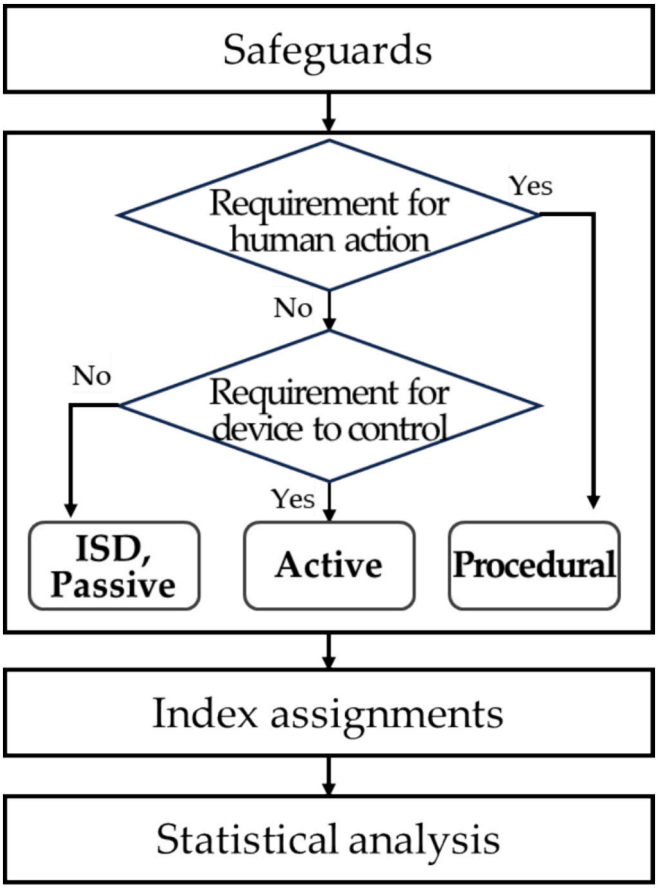


Fig. 7. Safeguard Evaluation Logic.

Table 3
Safeguard Categorization and Indexing.

Category	Index	Safeguard
Inherently safer design (ISD)/ Passive	A	Installation of new equipment for safety design (e.g., Non return valve (NRV))
	B	Modification of safety design
Active	C	Modification of control/valve system
	D	Installation of instrumentation devices
	E	Emergency system (e.g., Interlock)
	F	Alarm, warning systems, and monitoring system
	G	Pressure relief devices (PRVs)
Procedural	H	Clear labeling
	I	Thorough monitoring
	J	Physical inspection of equipment
	K	Adjustment of safety device settings
	L	Verification of proper operating procedures
	M	Communication among operators and supervisors
	N	Operator training

perspective. The results offer insights into the feasibility and limitations of fully LLM-based HAZOP automation.

3. Results

Building on the methodology outlined in Section 2, this section presents the results of evaluating LLM performance in generating HAZOP worksheets. In order to further validate the methodology during scenario identification, results from each model were randomly sampled and manually compared with reference HAZOP worksheets (See Appendix B). The following section reports the detailed results for each aspect.

3.1. Evaluation of model performance

3.1.1. Similarity results – model performance metric 1

Semantic and contextual similarity between LLM-generated worksheets and the expert-prepared reference worksheet was evaluated using BERT-based similarity scores, as summarized in Table 4. Results show that GPT4o achieved the highest F1 score (86.7 %), followed closely by GPT4o-mini and Gemini (both at 86.5 %), and LLAMA (86.2 %). Notably, LLAMA exhibited slightly lower precision (85.4 %), which reflects the generation of content not explicitly included in the reference worksheet. However, given its strong recall, LLAMA's output may contain both extraneous information and additional scenarios or safeguards that extend beyond expert-documented cases. While this tendency lowers measured precision, it still preserves contextual relevance and highlights LLAMA's potential for exploratory HAZOP. Overall, despite these differences, all four LLMs demonstrated substantial alignment with the reference worksheet, which confirm their capability to generate HAZOP worksheets with high semantic and contextual fidelity.

3.1.2. Cost results – model performance metric 2

The cost-efficiency of deploying LLMs for automated HAZOP worksheet generation was assessed by considering each model's pricing structure—including token-based input and output token costs, image processing charges—and the training data cutoff date, which influenced the model's relevance and reliability for real-time industrial applications.

Table 5 summarizes the training data cutoff dates and pricing structures for the four models assessed: GPT4o, GPT4o-mini, LLAMA 3.2, and Gemini 2.0. Each model was prompted with a HAZOP input consisting of 1,262 tokens. For consistency across evaluations, output was standardized to a maximum of 4,096 tokens per worksheet. Additionally, for models supporting image input, the cost of processing a high-resolution P&ID image (see Fig. 3, resolution: 1199 × 946 px) was included in the total cost estimate.

The results revealed notable differences in cost across the models. LLAMA 3.2 was the most cost-effective at only $0.011 per worksheet. As an open-source model, it does not require API licensing fees. However, its 90 billion parameters demand substantial computational resources. For this study, hardware-related costs were approximated using Microsoft Azure's GPU-enabled virtual machines (Azure Marketplace, 2025). Among commercial models, GPT4o-mini achieved the lowest cost, combining minimal tokens charges with moderate image processing fees. Its low per-task expense makes it attractive for large-scale or repetitive HAZOP analyses, in particularly budget-constrained settings. By contrast, Gemini 2.0 incurred the highest cost due to its advanced multimodal processing capabilities, including text-image interpretation. While this makes Gemini highly effective for complex, context-rich tasks, the elevated expense can limit its adoption in cos-sensitive industrial environments. GPT4o ranked in the mid-range, offering a balance between cost, training generative performance—making it a reasonable option for users prioritizing up-to-date knowledge while maintaining cost control.

Table 4
BERT-Based Similarity Scores for different LLMs.

Model	BERT Precision (%)	BERT Recall (%)	BERT F1 Score (%)
GPT4o	86.1	87.3	86.7
GPT4o-mini	86.1	87.0	86.5
LLAMA 3.2	85.4	87.0	86.2
Gemini 2.0	86.4	86.6	86.5

Table 5
LLM Training Data Recency and Cost Structure for Automated HAZOP Generation.

Model	Cutoff Date	Text Input Cost ($ / 1,000 tokens)	Text Output Cost ($ / 1,000 tokens)	Image Input Cost ($ / 150 × 150 px)	Total Cost per HAZOP Worksheet ($)
GPT4o	2024.06.	0.00250	0.01000	0.000638	0.076
GPT4o-mini	2023.10.	0.00015	0.00060	0.001275	0.067
LLAMA 3.2	2023.12.	0.00204	0.00204	0	0.011
Gemini 2.0	2024.08.	0.00010	0.00040	0.002250	0.115

3.2. Evaluation of HAZOP performance

3.2.1. Scenario results – HAZOP performance metric 1

Scenario identification is a core function of HAZOP, and in this study two indicators—the total number of generated scenarios and the valid scenario ratio—were used to evaluate this functionality. These metrics provide a quantitative basis for assessing each model's ability to independently identify meaningful incident scenarios. The results, illustrated in Fig. 8 (scenarios generation count), Fig. 9 (PCA-based clustering), and Fig. 10 (elbow method for cluster optimization), highlight both the generative capacity and the degree of alignment between model outputs and expert-prepared HAZOP data.

Scenario generation capacity varied significantly across the models, as shown in Fig. 8, which compares scenario counts at temperature settings of 0.1, 0.5, and 1.0. Gemini 2.0 demonstrated the largest number of scenarios, averaging 34.3 per worksheet for a total of 1,029 total across 30 runs. LLAMA 3.2 followed with 672 scenarios (22.4 per worksheet), while GPT4o and GPT4o-mini generated scenarios 608 (20.3 per worksheet) and 491 scenarios (16.4 per worksheet), respectively. As the reference HAZOP worksheet contained 25 scenarios, only Gemini exceeded this benchmark. The other three models, while generating substantial outputs, fell short of this reference value, indicating current limitations in fully automating the HAZOP process.

Scenario diversity across temperature setting revealed further differences among the models. Higher temperature settings are often assumed to promote diversity in generated outputs by increasing randomness. For example, LLAMA 3.2 displayed the wider interquartile range at the highest temperature seating (1.0) in Fig. 8, indicating greater variability in scenario generation. However, this effect was not consistently. GPT4o and GPT4o-mini produced consistent scenario counts across temperature variations, suggesting limited sensitivity to this parameter, while Gemini demonstrated the most consistent performance at a moderate temperature (0.5). Overall, these results indicate that temperature adjustments primarily affect surface-level linguistic diversity—such as word choice and phrasing—rather han the deeper logical reasoning required for structured scenario development. This interpretation aligns with Peeperkorn et al. (2024), who concluded that temperature tuning alone cannot explain or enhance LLM performance in structured engineering tasks like HAZOP. Given that a single, broad prompt was used to generate each complete HAZOP worksheet, it is likely that the Softmax-driven temperature effects were limited to stylistic aspects rather than impacting the structured reasoning required for scenario development. To improve reliability and consistency, future research should consider decomposing the HAZOP task into modular steps—such as node identification, deviation analysis, consequence evaluation, and safeguard suggestion—each governed by dedicated prompts and optimized temperature settings.

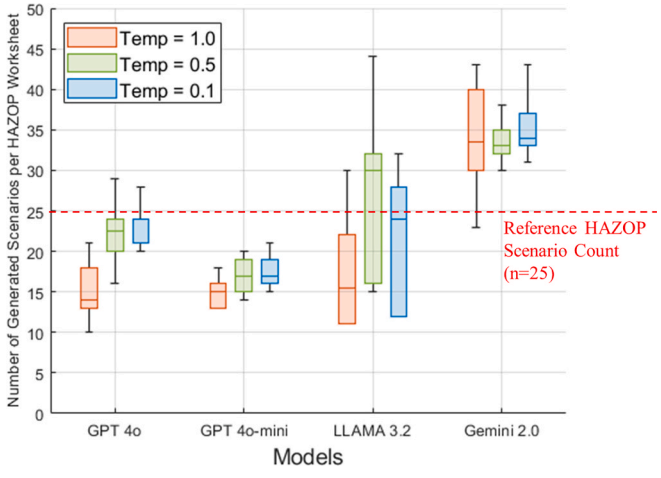


Fig. 8. Scenario Generation Comparison Across LLMs at Varying Temperatures.

Following the scenario count analysis, scenario validity was assessed using clustering-based analysis. High-dimensional BERT embeddings of the generated scenarios were first reduced to two principal components using PCA (Fig. 9), which preserved the most meaningful sematic variance while simplifying the structure for visualization and clustering. To identify the optimal number of clusters, the elbow method was employed, consistently indicating that three cluster ($k = 3$). This framework enabled a structured evaluation of how closely LLM-generated scenarios aligned with expert-prepared benchmarks.

The clustering results revealed clear differences in semantic alignment across the models. In Fig. 9, red points represent expert-generated baseline data from the reference worksheet, with the cluster containing the majority of these points defined as the "valid scenario cluster". GPT4o (Fig. 9a) showed distinct cluster boundaries, with a sizable portion of baseline data grouped within a single cluster, indicating moderate semantic alignment and structured scenario generation. GPT4o-mini (Fig. 9b) exhibited a similar pattern, with most baseline points residing in the indigo-colored cluster. This reflects high semantic accuracy with minor variability, likely due to paraphrasing or lexical differences. This suggests high semantic accuracy with minor variation, potentially due to differences in language structure or paraphrasing. LLAMA 3.2 (Fig. 9c) exhibited the most dispersed clustering pattern, with baseline data spread across the indigo-and gold-colored clusters. This suggests weaker alignment with expert data, indicating that although LLAMA generates a high volume of scenarios, they are semantically less consistent and thus lower in validity. In contrast, Gemini 2.0 (Fig. 9d) demonstrated the strongest coherence, with expert baselines concentrated in a single cluster and only limited overlap at cluster boundaries.

The computed valid scenario ratios further clarified these trends using Eq. (7). GPT4o achieved the highest alignment with expert benchmarks (37 %), followed by Gemini (33 %), GPT4o-mini (20 %), and LLAMA (19 %). These results confirm GPT4o strength in producing semantically valid outputs. Despite generating the largest number of scenarios, Gemini demonstrated comparatively solid alignment with expert benchmarks, highlighting its efficiency in producing high-quality outputs. Meanwhile, although LLAMA ranked second in scenario quantity, its low valid scenario ratio illustrates the trade-off between volume and relevance. GPT4o-mini generated the fewest scenarios, and its overall validity remains questionable.

In addition, a subset of data samples was manually compared with the reference HAZOP worksheet to validate the clustering results. While k-means clustering offers advantages such as computational efficiency and interpretability, it is highly sensitive to outliers; therefore, detailed examination within clusters is essential (Gao et al., 2023). Accordingly, ten scenarios were randomly selected from each of the three colored clusters generated by four models, resulting in 120 incident scenarios in total (Appendix B). The selection was carried out using the random number generation function in EXCEL to ensure unbiased sampling. These were then manually matched against the reference HAZOP

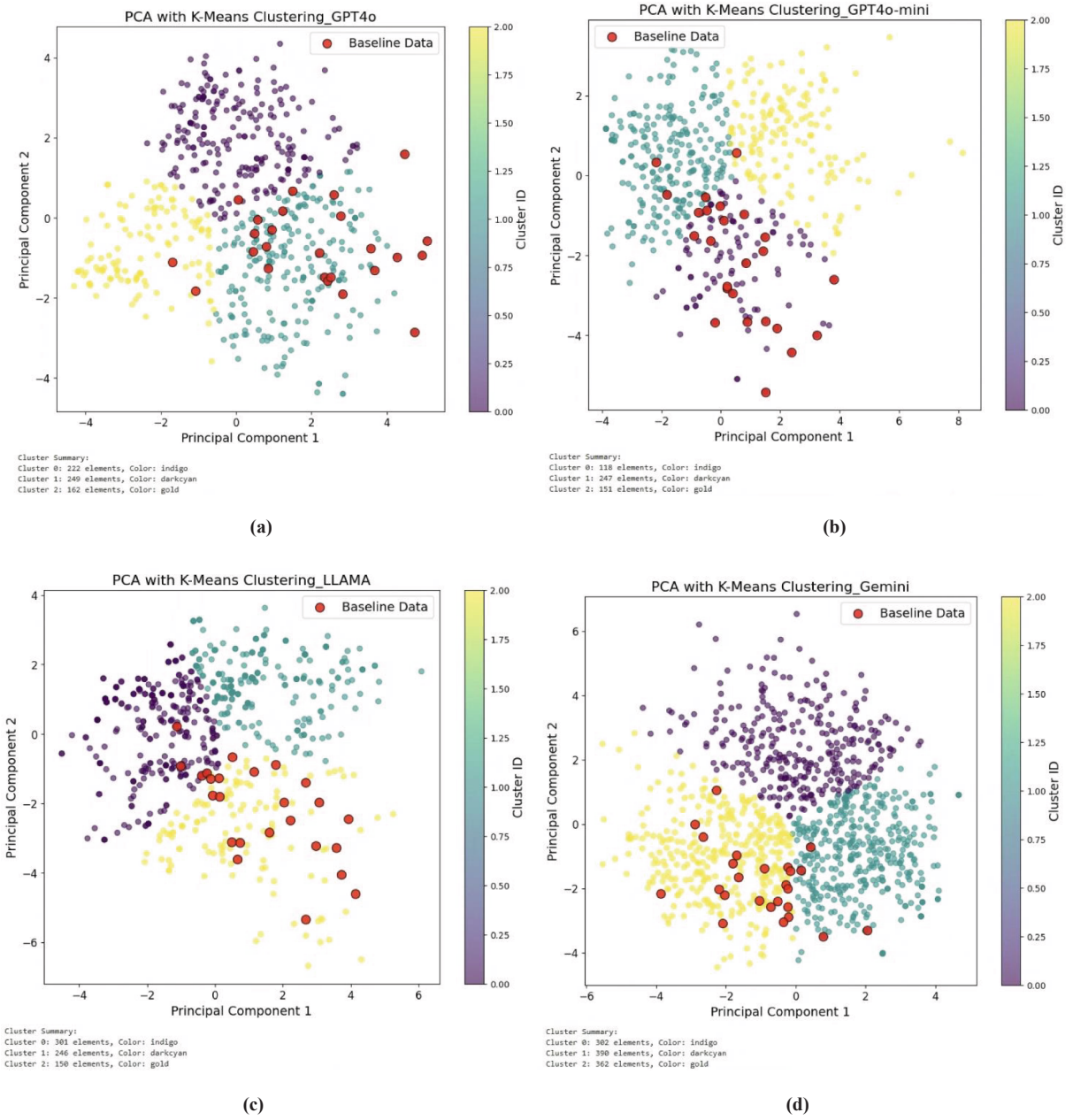


Fig. 9. PCA. (a) GPT 4o, (b) GPT4o-mini, (c) LLAMA 3.2, and (d) Gemini 2.0.

worksheet by the authors. As summarized in Table 6, the valid scenario ratios were 43 % for GPT-4o, 37 % for both Gemini and LLAMA, and 27 % for GPT-4o-mini. These findings are consistent with the k-means results and indicate that all four models face substantial challenges in generating incident scenarios.

A closer inspection revealed that many scenarios within semantically valid scenario cluster were all aligned with the reference HAZOP worksheet. The expert group primarily identified overpressure and C6 residue-induced leaks and fires as key consequences, and valid scenarios reflected these outcomes. Nonetheless, clear weaknesses were observed in terms of completeness and diversity. Several invalid cases within the valid clusters failed to match due to unclear object naming or ambiguous causal reasoning, and notably, four of GPT-4o's six valid scenarios were duplicates.

However, not all invalid scenarios were devoid of value. For example, LLAMA produced a lower proportion of valid scenarios within valid scenario clusters compared to the other three models. This was largely due to its focus on depressurization, rather than overpressure, as the primary consequence. Because the expert group did not consider depressurization as a consequence in their worksheet, these scenarios were categorized as invalid; nevertheless, under different criteria, they could be regarded as meaningful incident scenarios. Similarly, other model scenarios such as valve malfunctions or inadequate equipment installation, which were not included in the expert worksheet but could be considered plausible incident cases.

Additional valid scenarios were identified in the other two clusters, primarily resulting from repetitive communication errors or simple incident patterns, such as overpressure caused by excessive inert gas inflow. The invalid scenarios from these clusters could be broadly classified into three categories: (1) failure of object recognition, leading to missing or misconnected descriptions of key equipment such as valves; (2) superficial scenarios lacking causal linkages among deviation, cause, and consequence; and (3) descriptions of unrelated systems (e.g., blending), which showed little relevance to the reference HAZOP

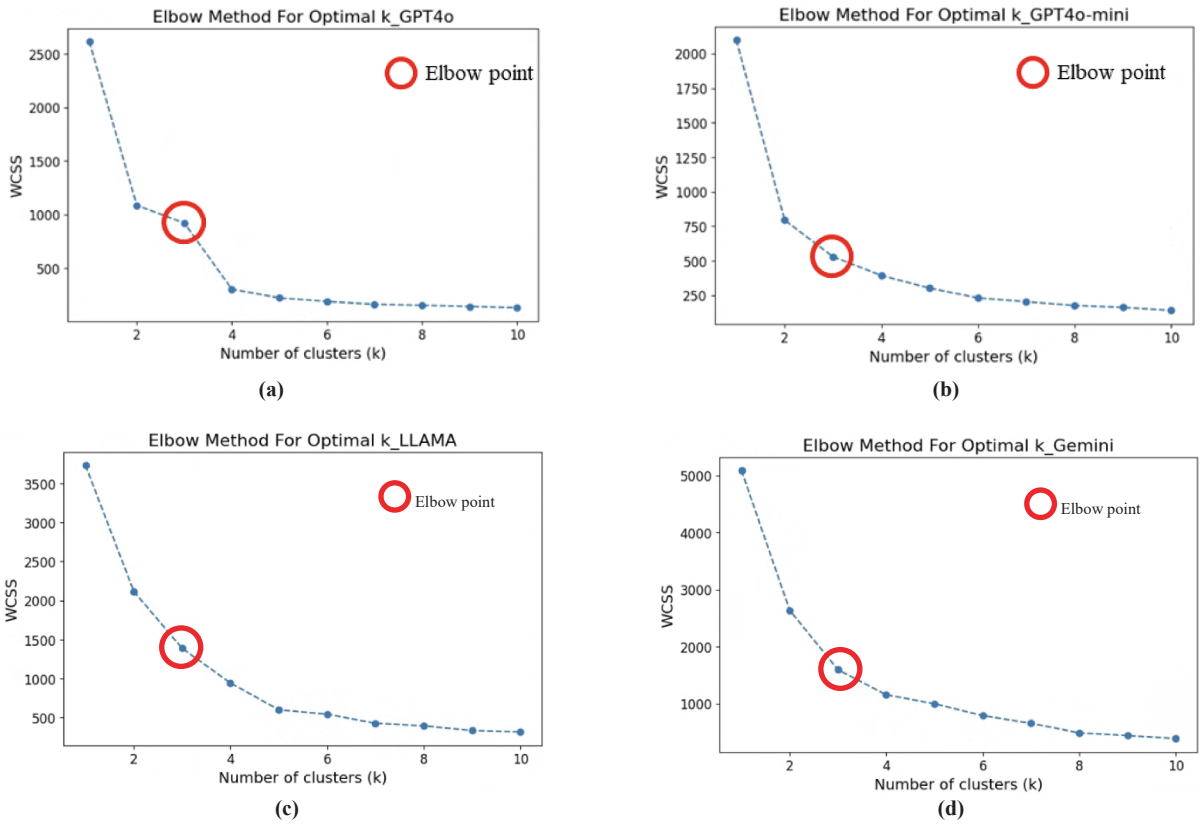


Fig. 10. Elbow method plots for determining the optimal number of clusters. (a) GPT4o, (b) GPT4o-mini, (c) LLAMA 3.2, and (d) Gemini 2.0.

Table 6
Valid Scenario Ratio from Manual Clustering Validation.

Model	Cluster	Number of valid scenarios	Valid scenario ratio	Total valid scenario ratio	K-means clustering results
GPT4o	Indigo	3	0.3	0.43	0.37
	Dark cyan*	6	0.6		
	Gold	4	0.4		
GPT4o-mini	Indigo*	4	0.4	0.27	0.20
	Dark cyan	3	0.3		
	Gold	1	0.1		
LLAMA	Indigo	5	0.5	0.37	0.19
	Dark cyan	5	0.5		
	Gold*	1	0.1		
Gemini	Indigo	1	0.1	0.33	0.33
	Dark cyan	4	0.4		
	Gold*	5	0.5		

* Valid scenario cluster.

worksheet. Ultimately, such scenarios were deemed meaningless, reflecting a structural limitation of LLM-based scenario generation.

3.2.1.1. Safeguard Results-HAZOP performance metric 2. The second performance metric evaluates the quality and characteristics of safeguards proposed by each language model. Table 7 summarizes the number of safeguards generated per HAZOP worksheet and per scenario. All four LLMs generated more safeguards compared to the reference HAZOP worksheet. Among them, Gemini 2.0 produced the highest number, averaging 61.1 safeguards per worksheet—more than three times the 20 safeguards listed in reference document. This outcome is consistent with Gemini's high scenario generation count. GPT4o, GPT4o-mini, and LLAMA 3.2 also outperformed the reference benchmark, producing between 24 and 31 safeguards per worksheet. On a per-scenario basis, Gemini 2.0 again led with 1.78 safeguards per scenario, followed by GPT4o, GPT4o-mini, and LLMA 3.2, while the reference worksheet offered 0.8 safeguards per scenario, the lowest number of safeguards. When normalized by the number of deviations, Gemini again ranked the highest with 1.78 safeguards per deviation, followed by GPT4o (1.53), GPT4o-mini (1.47), and LLAMA 3.2 (1.09). In comparison, the reference worksheet recorded just 0.8 safeguards per deviation, the lowest ratio among all models. This suggests that LLMs tend to propose multiple safeguards for each identified process deviation, potentially increasing the breadth of risk mitigation measures.

Table 7
Safeguard Production Performance by Model.

Criteria	Reference Worksheet	GPT4o	GPT4o-mini	LLAMA 3.2	Gemini 2.0
Number of safeguards / worksheets	20	31.0	24.1	24.4	61.1
Number of safeguards / deviations	0.8	1.53	1.47	1.09	1.78

Fig. 11 visualizes the distribution of safeguard types—categorized into passive, active, and procedural—across all models and the reference worksheet. The reference worksheet presents a relatively balanced safeguard profile, with 5 % passive, 30 % active, and 65 % procedural safeguards. In contrast, all LLMs displayed a strong preference for procedural safeguards. Gemini, for example, included 80 % procedural safeguards, with only 17 % active and 3 % passive. This procedural dominance was consistently observed across GPT4o, GPT4o-mini, and LLAMA. This overrepresentation of procedural safeguards suggests that LLMs tend to favor text-based, easily generated recommendations—such

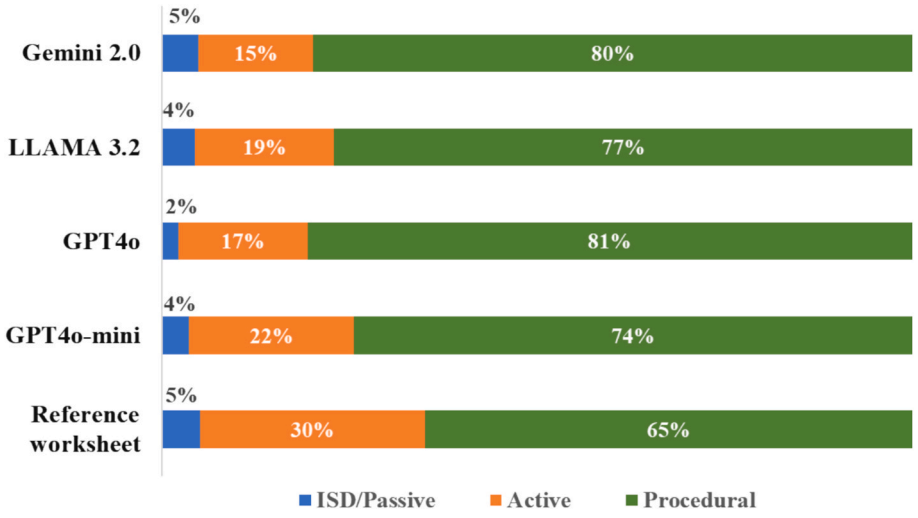


Fig. 11. Distribution of Safeguard Types Across Model: ISD/Passive, Active, and Procedural.

as verifying procedures or monitoring conditions—over more technically involved suggestions that require domain expertise, such as hardware modifications or control logic design. One likely explanation lies in the training data. While the exact datasets used to train these models are proprietary, it is reasonable to infer that much of the publicly available HAZOP-related content online emphasizes procedural recommendations. As a result, LLMs may have learned to disproportionately focus on procedural safeguards during training, contributing to the underrepresentation of passive and active control strategies.

To further examine safeguard quality, Fig. 12 presents a heatmap summarizing the safeguard characteristics generated by each model, categorized by the indexing system outlined in Table 3. Each row corresponds to a specific type of safeguard—ranging from inherently safer design (ISD) and passive measures (indices A–B), to active (C–G), and procedural safeguards (H–N). The color intensity represents the relative frequency of each safeguard type, with darker shades indicating more frequent occurrences. The reference worksheet shows a strong emphasis on a few specific safeguards, particularly Index G (pressure relief devices) and procedural elements like Index H (clear labeling) and Index N (operator training). While these represent foundational components in traditional HAZOP practices, they are by no means exhaustive. In contrast, the LLMs—especially GPT4o-mini, LLAMA, and Gemini—generated a broader spread of safeguard types, including several not emphasized in the reference worksheet. For instance, Gemini and GPT4o produced a wider range of procedural safeguards, such as Index I (monitoring) and Index L (verification of procedures), while GPT4o-mini offered frequent recommendations related to Index M (operator communication). Although procedural suggestions dominate across all LLMs, they appear in a more varied and distributed manner than those in the reference worksheet. This broader distribution might reflect the diversity of training data, and the general-purpose knowledge encoded in the models. When it comes to active safeguards, LLM outputs were more limited. Most models showed only marginal emphasis on safeguards like alarm systems (Index F) or control devices (Index C), and almost none recommended pressure relief devices (Index G) to the same extent as the reference. ISD and passive strategies (Indices A–B) remained largely underrepresented across all models and the reference worksheet alike—highlighting a broader trend of underutilization of inherently safer design principles in both human- and AI-generated outputs.

It is important to acknowledge that the reference worksheet itself may not be entirely comprehensive or fully optimized. From this

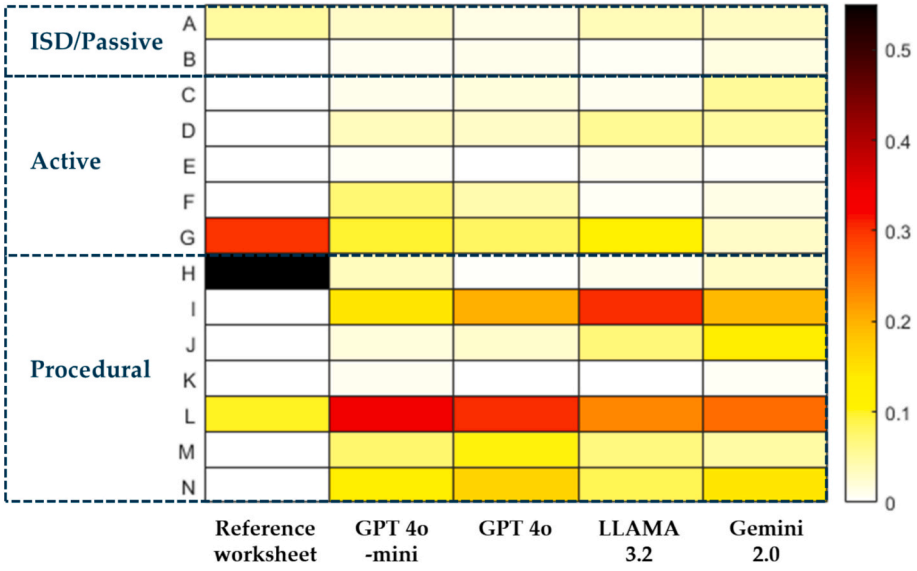


Fig. 12. Heatmap of Safeguard Characteristics by Each Model.

perspective, the diversity observed in LLM can be considered as a complementary strength rather than a limitation. The broader distribution of procedural strategies, along with alternative suggestions not included in the reference, indicates that LLMs can contribute to broadening the scope of considerations during early stage of HAZOP brainstorming.

4. Discussion

4.1. Answers to the research questions

Table 8 provides a consolidated analysis of the findings from Section 3 and summarizes the answers to the research questions presented in Section 1.

First, the answer to RQ1 is related to the similarity results. All four LLMs achieved BERT F1 Scores above 85 %, indicating that the structure and word composition of the generated worksheets resemble those of the reference worksheet. This suggests that LLMs are able to combine process parameters with guidewords to generate deviations, causes, consequences, and safeguards in a manner consistent with the HAZOP procedure. However, this similarity should not be overinterpreted. Because BERT F1 Score evaluates embeddings rather than natural language meaning directly, the results may mask contextual mismatches and semantic inaccuracies. In addition, it remains unclear whether the LLMs followed a reliable sequence of HAZOP steps when producing the worksheets, since their internal reasoning process is not observable. Therefore, these findings should not be taken as evidence of successful replication. Instead, they highlight the need for future studies to apply a chain-of-thought approach, prompting LLMs step-by-step through the HAZOP workflow to better evaluate their performance at each stage.

Second, RQ2 addresses the necessity of introducing LLMs into HAZOP, and the cost results provide the answer. The use of LLMs offers overwhelming advantages in terms of both cost and time. Specifically, the adoption of LLMs can drastically reduce the expenses associated with employing human experts and the long hours traditionally required to complete a HAZOP study. These strong advantages explain why the integration of LLMs into HAZOP strongly supports their consideration.

Third, RQ3 concerns the practical applicability of the HAZOP results. Based on the evaluation of HAZOP performance, the answer is negative. LLMs were able to generate a number of incident scenarios comparable to those in the reference worksheet, and even suggested one to three times as many safeguards. In traditional HAZOP, such output would typically require lengthy brainstorming sessions by human experts. However, closer inspection reveals clear limitations. The scenarios generated by LLMs diverge substantially from those in the reference worksheet, and the safeguards are heavily skewed toward procedural safeguards. Although a small fraction of these outputs may provide valuable insights overlooked by humans, such cases appear to be rare. Therefore, the results produced by LLMs require prior expert review, and their standalone practical applicability should be regarded as limited.

Finally, the results of RQ1 through RQ3 allow us to address RQ4: What role should current LLMs play in HAZOP—fully automation or supportive tool—and what are the implications for future development? The findings indicate that while LLMs show a potential feasibility for automating HAZOP, at present their role should be regarded primarily as supportive tools that require careful expert validation. For LLMs to gain a more reliable footing in the HAZOP domain, efforts must focus on improving the transparency of how scenarios and safeguards are generated. A step-by-step prompting strategy, in which each stage of the HAZOP workflow is explicitly guided, is recommended. Furthermore, the development of quality indices for evaluating generated scenarios and safeguards is needed. By leveraging operational records or incident investigation reports to train or fine-tune LLMs with these indices as benchmarks, users may enhance both the quality and reliability of the outputs.

Table 8
Summary of findings for research questions (RQ1-RQ3).

Research Question		Findings
RQ1	Can current LLMs perform the entire HAZOP procedure autonomously?	LLMs are able to generate worksheets that follow the HAZOP procedure and resemble the reference worksheets. However, it does not necessarily imply that the generated worksheets are accurate or practically meaningful.
RQ2	Does LLM-based HAZOP provide advantages in terms of time and cost compared to traditional methods?	In terms of time and cost, LLM-based HAZOP offers clear advantages over traditional methods.
RQ3	Is the quality of HAZOP results generated by LLMs sufficient for practical application?	The limitations are evident. The proportion of valid scenarios generated is small, and the safeguards are heavily biased toward procedural measures, raising concerns regarding their practical applicability.

4.2. Threats to validity

4.2.1. Construct validity

Construct validity concerns whether the chosen measurement approaches reliably capture the intended research outcomes (Runeson, et al., 2012). In this study, assessing what constitutes a "well-conducted" HAZOP—particularly in relation to RQ1(feasibility) and RQ3 (practical applicability)—inevitably required expert judgment, which introduces the risk of subjective bias. To mitigate this threat, two complementary validation methods were employed: the BERT F1 Score and k-means clustering. The BERT F1 Score for RQ1, widely used in NLP research, measures textual similarity and was employed to assess the alignment between reference HAZOP worksheets and those generated by LLMs. However, as a "similarity"-based metric, it does not account for the causal relationships unique to HAZOP— linking deviations, causes, consequences, and safeguards—and thus may overestimate their quality. Likewise, unsupervised k-means clustering was applied to evaluate RQ3, but its accuracy can vary depending on domain-specific terminology and phrasing (Zhao, et al., 2025). Recognizing these limitations, representative samples from each cluster were further examined to validate the robustness of the results.

4.2.2. Internal validity

Internal validity addresses potential risks in how results are obtained (Runeson et al., 2012). In this study, zero-shot prompting was adopted to facilitate a straightforward comparison of LLMs, and performance variations resulting from prompt engineering were not considered. In other words, the models were not allowed to reattempt answers or refine incomplete responses. This limitation implies that the evaluation of LLM's feasibility and practical applicability in HAZOP was conducted without reflecting the models' sensitivity to different prompting strategies, a factor known to significantly influence performance. Future studies should therefore redesign prompts in a structured pipeline (node identification, deviation, cause, consequences and safeguards) and incorporate fine-tuning with chemical incident databases. Such approaches would provide a more accurate assessment of LLM performance in optimized HAZOP applications.

4.2.3. External validity

External validity relates to the generalizability of research findings (Runeson et al., 2012). This study relied on only a single reference HAZOP worksheet, which was derived from a textbook that may be simpler, less complete, and potentially contain errors compared to HAZOP results conducted on actual industrial systems. Therefore, extending the evaluation of RQ1 (feasibility) and RQ3 (practical applicability) to more complex process diagrams and practitioner-generated

HAZOPs is a critical step for enhancing generalizability. However, most of these datasets are classified as highly confidential, making their external release and sharing restricted. Moreover, industries are particularly cautious about allowing such proprietary data to be used for LLM training. As a result, the application of real industrial HAZOP data in this study was limited. To address this external threat, the study instead employed guideline-based examples disseminated by authoritative institutions and provided detailed contextualization of the research questions, ensuring that the analysis remained focused despite these data constraints. Accordingly, the findings should not be generalized beyond the scope of the defined RQs, and future validation across diverse systems is necessary.

Another external threat arises from the variability of LLM outputs, which raises concerns about reproducibility and reliability. Because LLMs can generate different worksheets for the same prompt, consistency is not guaranteed. To mitigate this issue, 30 worksheets were generated per model, with the system reset before each run to minimize memory bias. These measures reduced, though did not eliminated, the threat of inconsistency in the study's results.

5. Conclusion

This study examined whether recent multimodal large language models (LLMs)—specifically GPT4o, GPT4o-mini, LLAMA 3.2, and Gemini 2.0—can independently perform a complete Hazard and Operability Study (HAZOP) without human intervention. A total of 120 HAZOP worksheets were generated from a single standardized prompt using each model, and their outputs were benchmarked against a reference expert-prepared worksheet. Instead of a direct item-by-item comparison, the evaluation was framed in terms of two perspectives: overall model performance and HAZOP performance. Model performance was assessed through semantic similarity (BERT F1 score) and cost efficiency, while HAZOP performance was evaluated through scenario generation capacity and safeguard diversity.

The findings indicate that all four models achieved high semantic similarity to the reference worksheet, thereby confirming the feasibility of LLMs (RQ1) in producing syntactically and semantically appropriate outputs at substantially lower cost of traditional expert-led processes, thus demonstrating their cost efficiency (RQ2). However, limitations became evident in HAZOP execution. Although Gemini 2.0 generated 1.4 times more scenarios than the reference, and the other three models produced comparable volumes, k-means clustering and manual analysis through sample extraction revealed that the valid scenario ratio across models remained only 19–37 %. This underscores the difficulty of attaining reliable scenario quality. Similarly, while all models proposed more safeguards than the reference (up to three times more in the case of Gemini), the majority were procedural in nature. This bias toward procedural controls indicates a limited scope of safeguard reasoning, with active and design-level interventions—critical for robust process safety—largely underrepresented. Consequently, the practical applicability of LLMs to HAZOP (RQ3) is remains markedly limited, suggesting that expert involvement is indispensable and that, at present, LLMs are best positioned as auxiliary tools rather than autonomous agents in the HAZOP process.

To address these challenges, future research should prioritize three directions. First, prompt engineering must evolve into a more structured methodology. Rather than issuing a single prompt, the HAZOP task should be decomposed into modular components—node identification, deviation analysis, consequence assessment, and safeguard proposal—each guided by targeted prompts. This approach would reduce ambiguity, enhance consistency, and improve semantic accuracy. Second, fine-tuning LLMs with domain-specific HAZOP datasets could significantly improve their capacity to recognize and recommend design-based and inherently safer interventions. Current models show clear potential in producing procedural guidance but struggle with generating more technically nuanced outputs. Third, post-processing and expert-in-the-loop frameworks should be explored to combine LLM efficiency with human oversight. Such hybrid model can capitalize on the strengths of LLMs—namely speed, breadth, and idea generation—while mitigating their weaknesses through expert validation and correction.

In conclusion, while none of the evaluated LLMs are currently capable of fully replacing expert involvement in the HAZOP process, their utility as supportive tools is evident. With minimal cost and setup, LLMs can rapidly generate preliminary drafts of HAZOP worksheets that may serve as a foundation for expert review. This accelerates early-stage analysis, reduces resource burdens, and fosters ideation. As model architectures improve and domain-specific training advances, the potential for partially or fully automated HAZOP processes will continue to grow. For now, LLMs should be regarded not as replacements for human judgment but as powerful tools that augment and enhance safety engineering practices.

CRediT authorship contribution statement

Junseo Lee: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Sunhwa Park:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition. **Sehyeon Oh:** Writing – review & editing, Methodology. **Byungchol Ma:** Writing – review & editing, Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

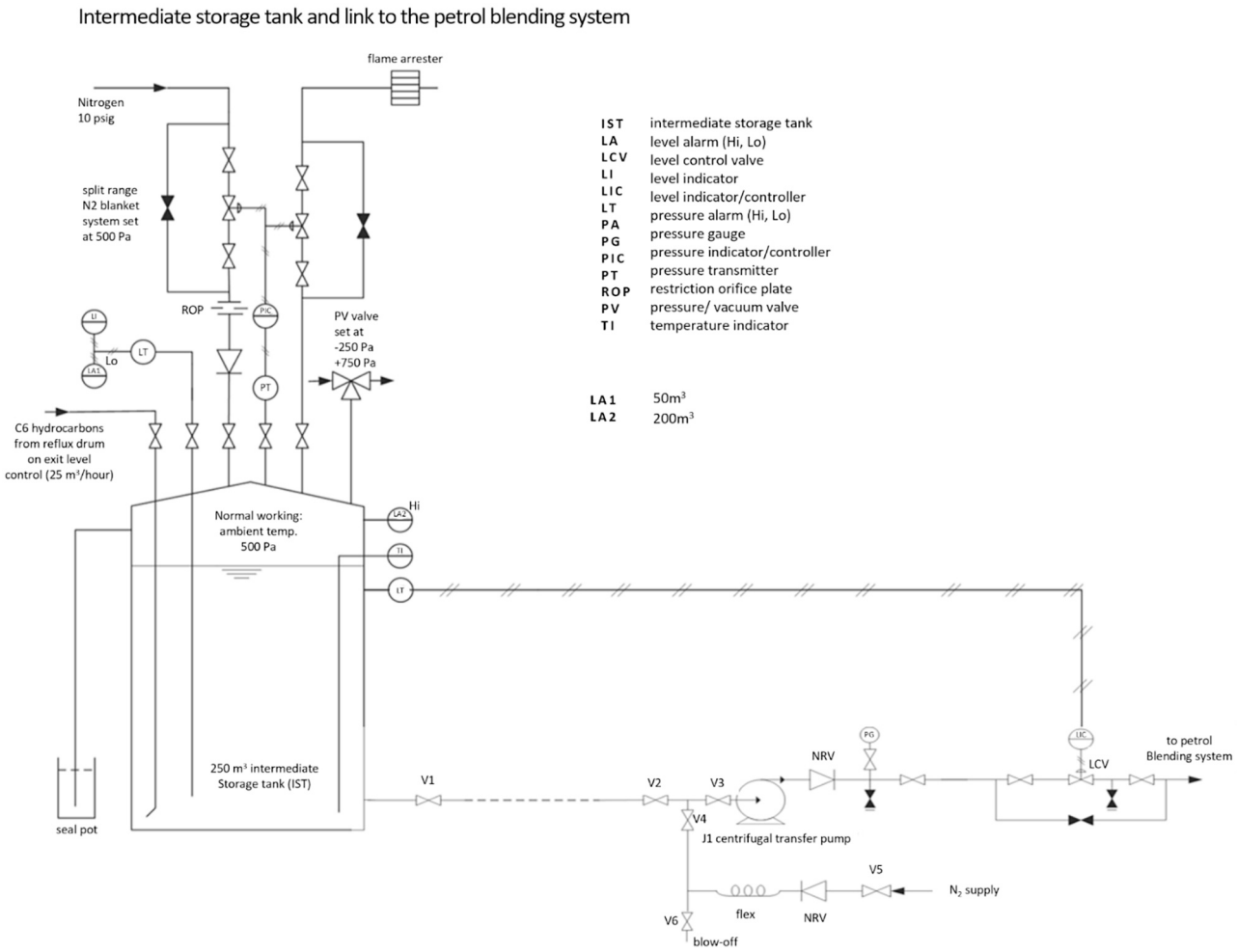
Appendix A

Input data

Prompt

"Read the P&ID image, perform the HAZOP by referring to the process diagrams in A.1 and A.2, and the HAZOP guidelines in A.3. And then prepare a table-style HAZOP Sheet for each node as presented in A.3. The HAZOP sheet should be written in the order of Guideword/Effect/Cause/Consequence/Safeguard."

P&ID



Background

This example is based on the HAZOP study of a planned modification to an existing process operation. The intermediate storage tank (IST) in question receives a C6 hydrocarbon stream, averaging 25 m^3/hour, from the reflux drum of an atmospheric pressure distillation column. The stream is routed, under exit-level control, via reflux pumps into a 250 m^3, nitrogen-blanketed tank. This conical-roofed tank acts as a buffer and temporary storage for the material before the C6 hydrocarbon is pumped, under level control, by the J1 centrifugal pump to the plant petrol blending unit.

The IST operates at ambient temperature and is maintained at 500 Pa using split-range pressure control. It is inerted by nitrogen supplied from the site's 1.3 nitrogen system. The tank is protected by a pressure (vacuum) valve (PV) set at $-250/+750$ Pa. The tank is located within a bunded enclosure with an overflow, sealed with glycol, that empties into the bund. Adequate instrumentation is provided, including level indicators with high- and low-level alarms, a high-level trip, and temperature and pressure indicators—all connected to the site control room for monitoring.

A P&ID image provides sufficient detail for the procedural HAZOP, and any deficiencies in the design or documentation fall outside the scope of this study.

The proposed modification involves removing the 100-meter length of 100-mm diameter piping between the tank's master isolation valve (V1) and the first pump isolation valve (V2). The pump will be refitted closer to the tank. However, before the modification, the flammable fluid in the piping (approximately 0.8 m^3) must be removed. Initially, draining the fluid into a drum was considered, but the associated risks were deemed unacceptable. As part of the corporate management of change (MOC) policy, a hazard study (HS) approach was adopted.

1) Inherently safer option (HS 0): The decision was made to displace the flammable fluid using nitrogen. The site's nitrogen ring main can be connected below valve V4, with the connection fitted with a non-return valve at the hose connection point.
2) FEED stage (HS 2): During the front-end engineering design (FEED) stage, this was identified as a non-standard operation with potential human factor risks. It was recommended that all valves be clearly labeled (e.g., V1) and that a dummy-run practice be conducted to debug the procedure and familiarize the crew with the operation.
3) HAZOP Stage (HS 3): It was ultimately decided that a HAZOP study should be carried out to ensure thorough analysis.

The final arrangement includes a 100-mm diameter suction line, a 25-mm diameter nitrogen header, a flexible hose, and an 18-mm diameter depressurizing line with an isolation valve (V6). All piping, except for the hose, will be hard-piped.

An operational procedure has been drafted, which, in accordance with the MOC policy, will undergo a procedural HAZOP to identify and mitigate potential risks.

Detailed proposed sequence

The operation will be executed by two operators: the lead operator stationed near J1, responsible for controlling the procedure, and a second operator located at the tank to operate valve V1, maintaining radio communication throughout the process. The initial setup requires all valves (V1–V6) to be closed, with the line between V1 and V2 containing C6 liquid.

1) Open V6 then open V5 to prove the line clear of debris and to displace any air in the hose.
2) Close V6 then open V4.
3) The operator at the J1 pump should open V2 slowly until it is fully open.
4) The operator at the tank is then instructed to open V1 slowly by one or two turns.
5) The operator at V1 should wait until nitrogen is heard passing through the valve into the IST then the tank operator will close V1.
6) The tank operator should then cautiously reopen V1 by one or two turns to ensure as much liquid as possible has been blown back to the IST.
7) Close V4.
8) Close V1 after allowing any residual N2 in the line to de-pressure into IST.
9) Pump-based operator to close V2.
10) Close V5.
11) Verify V2, V4, and V5 are all closed.
12) Open V6 to depressurize the line.
13) Disconnect the hose at V4.

HAZOP study

Node 1. Steps 1–2 (Connect and prove the nitrogen supply: V5-V6)

Design intention: To ensure the nitrogen supply is correctly fitted and to displace any air present in the hose.

Node 2. Steps 3–9 (Clear the line by blowback to IST: V5-V1)

Design intention: To completely clear the petroleum feedstock from the 100-meter line between the J1 pump and the IST. This is achieved through blowback using nitrogen supplied from the 1.3 barg nitrogen ring main via a temporary connection fitted to an existing drain at the J1 pump. Manual control is maintained by operators stationed at each end of the line. After the main clearance, a brief second flush will be applied.

Node 3. Steps 10–13 (Depressurize and disconnect: V1-V6)

Design intention: To safely depressurize the line, which previously contained C6 and now contains nitrogen.

Node 4. V3-J1-LCV (Line for petrol blending)

Design intention: Not yet completed.

Guide words for HAZOP study

- Out of Sequence: Too early, too late
- Rate: Too fast, too slow
- Magnitude: More, less
- Pressure: More
- Communication: Ineffective communication between operators
- Reverse: Reverse flow Incomplete: Incomplete execution of steps
- Other: Additional deviations not covered by the above guidewords

The reference HAZOP worksheet

Guideword	Effect	Cause	Consequence	Safeguards
Out of sequence	No flow N_2	V6 closed, V5 open	Some (small amount) of air left in the hose	Valve labeling and practice
Rate Too fast	High flow N_2	V5 too far open	Waste of N_2 and local noiseJet reaction on stones at vent, V6	Valve labeling and practice
Out of sequence	Possible contamination of downstream process or rotation of the pump	V3 open by mistake, V6 closed, and V4 opened	Possible contamination of downstream process or rotation of the pump	Valve labeling and practice
Out of sequence/ reverse	Reverse flow from (upstream) process to V6	V3 left open, V4 left open, V6 open during the depressuring	Possible release of “materials” from downstream plant	Pump NRV Valve labeling and practice Many valves in route

(*continued on next page*)

(*continued*)

Guideword	Effect	Cause	Consequence	Safeguards
Out of sequence/ reverse	Reverse flow from IST	V6 left open after blowdown and V1 and V2 opened ready for displacement of C_6. V5 still closed	IST drains through V6Environmental impact Potential fire	Valve labeling and practice
Out of sequence Too early	N_2 flow out of V6	V6 left open	N_2 losses	Valve labeling and practice
Out of sequence Too early	Possible release of C6 at V6	Human factors. V6 opened and V1 and V2 opened for displacement. V5 not yet opened	Environmental impact and possible fire	Valve labeling and practice
Out of sequence Too late	N_2 in next (downstream) operation	V2 not open, V3 left open	None	
Out of sequence	C_6 released from V6	V1 and V2 open, V6 left open, and V5 closed	Environmental impact and possible fire	Valve labeling and practice
Rate Too fast	V5 too far open	Human factors	Possible overpressure of IST due to high N_2 flow	PRV on IST
Pressure More	V5 too far open	Poor understanding of operation	Possible overpressure of IST due to high N2 flow	PRV on IST
Magnitude	More N2 flow into IST	Poor understanding of operation	Possible overpressure of IST due to high N2 flow	PRV on IST
Rate Too slow	C_6 not displaced	No true indication of N_2 flow rate	Slower displacement of C_6 due to N2 "slippage"—wavy flow may result in limited C_6 removal	None
More flow N_2 High flow	V5 too far open	Human factors	Possible overpressure of IST due to high N_2 flow	PRV on IST
More flow N_2 High flow	V5 too far open	Poor understanding of operation	Possible overpressure of IST due to high N2 flow	PRV on IST
Incomplete	C_6 left in line	Line not true, hogs and hollows plus elevation changes	Some C_6 trapped in the line at the end of the final blow through. Environmental impact and possible fire	None obvious
Reverse	C_6 released from V6	V6 left open and V5 closed at the end of cycle. Some C_6 still in the line	Environmental impact and possible fire	Valve labeling and practice
Communication	C_6 released from V6	V6 left open and V5 closed at the end of cycle. Some C_6 still in the line	Environmental impact and possible fire	Valve labeling and practice
Communication	Misinterpretation	Human factorsWhat is the significance of a change in the noise?What will it sound like?	Wavy flow may produce a sound like gas passing into the IST.	Valve labeling and practice
Communication	Misinterpretation	Misunderstanding of the point in the sequence without a clear lead operatorMisunderstanding of the point in the operation due to poor radio protocol	Possible upset not easy to define	Lead operator is specified in the procedure
Communication	Possible source of ignition	Radios not compatible with Hazardous Area Classification	Possible fire (remote possibility)	Lead operator is specified in the procedure
Incomplete	Possible reverse flow from IST during the step 8 depressuring cycle	Hydrostatic head in IST	Live V1—V2 is decontaminated with C_6Possible environmental impact and fire during final blow down through V6	None
Less flow Incomplete	Line V1—V2 incompletely cleared. C_6 still in line	Flow regime uncertain and line slopes uncertain	Significant final C_6 left in line which has to be drained.Environmental impact and risk of fire	None
More flow (N_2) (V5 too far open) (See 2.3)	V5 too far open	Human factorsPoor understanding of operation	Possible overpressure of IST	PRV on IST
Communication	C_6 still in line. Could result in a major spill later in the process	"Sound" is the only variable. What will it "sound like"?	Environmental impact and possible fire if incompletely drained	None

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ssci.2025.107039.

Data availability

Data will be made available on request.

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